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A Study on Face Detection Using Viola-Jones Algorithm in Various Backgrounds, Angles and Distances

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Abstract: *In recent years, research in image processing is very popular and demanding because of its diverse and humongous application. Face detection is one of the dominant mechanisms for security purpose, video surveillance, humane computer interaction, etc. Numerous algorithms have been proposed by many types of research regarding face detection and Viola-Jones algorithm is one of them. This paper deals with Viola-Jones algorithm for face detection in various positions and surroundings which include distance from the camera, background color, the angle of the object, etc. Further, this paper also performs an analysis in order to find the best settings that are appropriate for Viola-Jones algorithm.*
Keywords: *Face detection, Viola-Jones algorithm, Various background*

1. Introduction

Human life is augmented by artificial intelligence, in which a machine exhibits intellectual functions. Computer vision tends to mimic humane vision [1]. For more than 20 years, research in this sector has been operated and therefore we can say that the current status of this field is progressive. Face detection is a very famous and attractive research topic since it has multifarious applications. One of the biometric authentication techniques is face recognition and face detection is the primary step of it [2]. Human beings are a gift with an amazing inborn potential to detect and recognize a person by just using his memory. Unfortunately, the computer does not have this function and that is where artificial intelligence appears.

Computer vision-based action recognition, face recognition, facial action unit recognition, eye gaze detection, tiredness understanding, emotion understanding, tracking, gait analysis, healthcare, etc. are very important active research areas [4-6,12,14,15,21,36]. Though recording 'face' relates to privacy issues, it is a very widely used scheme in numerous vision-based applications over a decade. By doing sophisticated programming and training, a machine can learn and duplicate humane activities. Visual emotion perception can be analyzed by face-scanning [3]. Face detection and then approaching various actions from detected faces are gaining in many applications [4,5,29,36].

The function of this face detector is to detect a face and track it wherever it goes where the input is taken from a video camera. Different approaches can be adhered [6]. An exact numerical description of faces and non-faces should be given to a computer in order to train the computer. These characteristics can be extracted with AdaBoost. It is an exceptional committee learning algorithm and uses the voting mechanism to form a strong classifier from the weak one [7]. Viola-Jones object detection algorithm was proposed by Viola and Jones in 2001 where techniques such as integral image and cascade make the algorithm highly efficient [8]. This algorithm first searches for vertical bright bands in an image which might be noses. Then it finds out horizontal dark bands which might be eyes. Finally, it looks for other general patterns associated with faces. These features individually may not suggest face strongly, but once these are detected one after another in a cascade, they are

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strongly suggestive of the face. As long as the average time to detect face is concerned, this algorithm shows better performance than other face recognition techniques, such as Gabor features extraction and project profile analysis [9].

In this research, we will use this Viola-Jones algorithm in various surroundings and environmental conditions. If we consider a video or an image of a person where the background is excluded, that would be unrealistic. Therefore, a more pragmatic approach would be to take a video in a complex background where other things are taken into account. The recognition rate will differ in every case.

The paper is organized as follows. Various methods of face detection are discussed in Section 2. Section 3 discusses Viola-Jones algorithm. The experimental setup, the detection result, and its analysis are shown in Section 4. Section 5 concludes the paper with a summary.

2. Literature Review

In this section, we will have a look at existing face detection techniques. 3D facial detection is getting better and can be classified into four categories. They are presented in the following sub-sections.

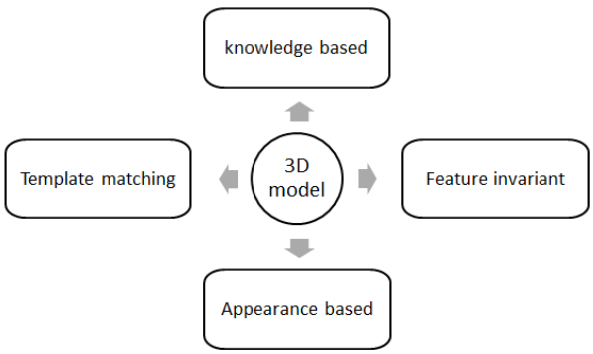


Figure 1. Various types of face-detection

2.1. Knowledge-based method

This method exploits the human knowledge of the typical geometry of human face and facial organization [10]. Face localization can be achieved by using these methods. Yang and Huang [11] used a knowledge-based method in a hierarchical form to detect faces. There are three levels of rules in their rules. At the highest level, the window method is used for finding all possible face candidates. A window is scanned over the input image and a set of rules is applied at each location. A rule-based localization was proposed by Kotropoulos and Pitas [31] where the horizontal and vertical profiles of an input image are obtained.

2.2. Feature invariant method

Contrary to the knowledge-based method, feature invariant method finds structural features for face detection. Different structural features like facial local features, shapes, texture and skin color are being used [10]. A statistical model is then developed to depict their associations to certify the presence of a face. Sirohey [13] proposed a method in which the head is separated from the background clutter by employing an elliptical model of head. Through some preprocessing, it utilizes this information present in the edge map and an ellipse is then fitted to mark the borderline between the head area and the background. Augusteijn and Skujca [32] developed a method that used face-like texture to detect the presence of a face. They examine skin, hair and other features of the face. Textures were computed by exploiting the second-order statistical features. Sub-images of 16 × 16 were taken for this purpose.

2.3. Template matching approaches

Template matching approach makes use of correlation values. Here, at first, a function is used for a standard face pattern. After an input image is given, the correlation values along with the standard patterns are a measure for different parts like eyes, nose, face contour and mouth [33]. A multi-resolution template matching with color segmentation and region clustering was developed by Ping et al. [34]. Jin et al. [17] also use a template matching approach but in a different manner. Firstly, skin color information is used based on a luminance-conditional distribution model; then, morphological operations are used and followed by template matching. Here, for detecting a face in each skin-region rectangle, template matching approach is based on linear transformation.

2.4. Appearance-based methods

The basic difference between the template matching method and appearance-based method is that the later approach can learn from a large number of examples [4]. Detecting faces are similar to a two-class pattern classification problem, which are face and non-face. All face images are in the face class and the non-face class contains images that may portray anything but a face [6].

principal component analysis (PCA) along with Eigenface technique is an example of an appearance-based method and one of the most used algorithms. Mutelo et al. [18] developed two-dimensional reductions PCA (2D PCA) which represents data image matrices within fewer rows and columns. Jee et al. [19] utilize support vector machine (SVM) for detecting face region which is a linear classifier. Anand and Lawrance [20] proposed a face detection and recognition algorithm based on the left-right hidden Markov models (HMM). It also uses singular value decomposition (SVD) coefficients. The human face is segmented into seven facial regions and for choosing facial features, a reduced set of quantized SVD coefficients was trained. Jameel [35] used HMM and PCA coefficients as features and then applied discrete cosine transform (DCT). It then uses seven-states HMM to model face configuration. Hashemi and Gharahbagh [22] proposed a method for feature extraction that uses wavelet transform with eigen value of 2D wavelet coefficients matrix. Here, radial basis function neural network is used as a classifier. Gao and Lee [23] developed a method combining the affine scale invariant feature transform (SIFT) and probabilistic similarity, for face detection and recognition under a series of different viewpoints. Merging of both global and local features with the use of PCA and local binary pattern (LBP) has been developed by Sompura and Gupta [24]. It results in one vector which is used by multilayer perceptron (MLP). Alshebani et al. [25] presented an innovative method for feature extraction. The features are extracted after segmenting the facial regions of the nose, eyes, and mouth manually. For calculation of the local representations of these regions, Gabor transform of the maximum of these areas is extracted. And for the classification, the K nearest neighbor (KNN) method is used. Wu [26] uses horizontal component prior principle (HCPP) and to improve LBP operator, it exploits the regional directional weighted LBP (RDW-LBP).

3. Viola-Jones Method

Viola-Jones algorithm is the most famous algorithm for face detection. It is a framework that is used in real time and the training rate is very high. There are three major ideas in Viola-Jones face detection method: the integral image, AdaBoost classifier learning, and finally the attentional cascade structure.

3.1. Rectangular feature

Instead of pixels, Viola-Jones algorithm uses rectangle features. A finite quantity of training data is insufficient to encode ad-hoc domain knowledge. Rectangular features can solve this problem [27].

In Figure 2, two-rectangle feature means the difference between the summations of the pixels inside two rectangular zones. Consequently, three-rectangle feature means the summation of two external rectangular regions subtracted from the summation in a center rectangle. Difference between diagonal pairs of rectangles is indicated by four-rectangle feature [8].

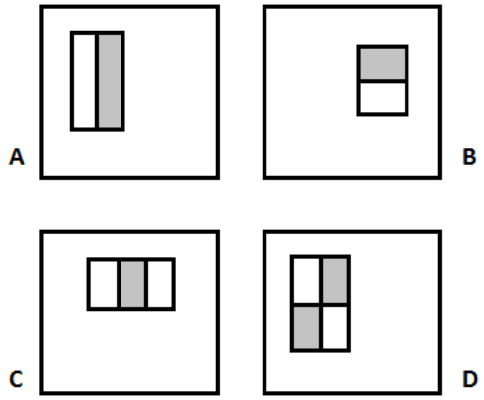


Figure 2. Rectangle features relative to enclosing detection window

3.2. Integral image

At locations x, y, the integral image holds the sum of the pixels above and to the left of x, y inclusive:

$$ii(x,y) = \sum_{x' \le x, y' \le y} i(x',y') \quad (1)$$

The computation of integral images can be carried out in one pass over the original image by utilizing the following pair of recurrences:

$$s(x,y) = s\,(x,y-1) + i(x,y) \quad (2)$$

$$ii(x,y) = ii(x-1,y) + s(x,y) \quad (3)$$

Here, $s(x,y)$ means the cumulative row sum.

In rectangle D, with four array references, the sum of the pixels can be computed. These are: the summation of the pixels in rectangle indicated as A is assigned as the value of the integral image at location 1. A+B is the value at location 2. The value at location 3 is A+C, and A+B+C+D is considered as the value at location 4.

Computation of sum within D is done as 4 + 1 − (2 + 3). For any scale, six array references are used to compute a two-rectangle feature [8].

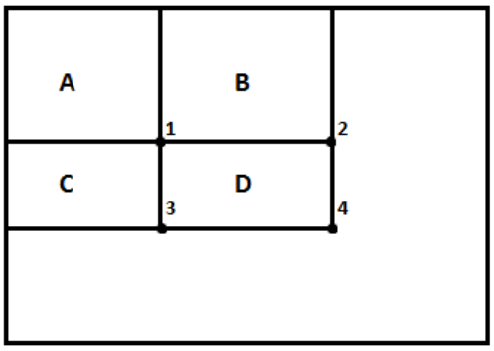


Figure 3. Demonstration of the sum of the pixels within the rectangle D – with four array references

3.3. Learning classification functions

Variant AdaBoost is used for feature selection and classifier training purpose [28]. It combines a group of weak classification functions and forms a classifier which is stronger. To boost the weak learner, it is to solve a series of learning problems. Once the first round of learning is completed, the examples are re-weighted. The previous weak classifier classifies with less accuracy and the re-weighting process takes care of those. The final classifier is a strong one and takes the structure of a perceptron [30].

3.4. AdaBoost for feature selection

Given example images x_i and labels $y_i = \{0,1\}$, let us initialize weights according to the following equation:

$$w_{i,1} = \frac{1}{2m}, \frac{1}{2l} \quad (4)$$

where m and l respectively mean the number of negative and positives. For $t = 1,\ldots,T$, we normalize the weights:

$$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}} \quad (5)$$

Afterward, we select the min-error classifier ht,

$$\epsilon_t = min_{f,p,\Theta} \sum_i w_i \mid h(x_i, f, p, \Theta) - y_i \mid \quad (6)$$

If x_i is classified incorrectly, its weight is not modified. Otherwise, its weight needs to be adjusted according to the following equation:

$$w_{t+1} = w_{t,i} \frac{\epsilon_t}{1-\epsilon_t} \quad (7)$$

3.5. Final (Strong) classifier

Weak classifiers are linearly combined and this combination forms the final classifier. Performance of each classifier is used to weigh it:

$$C(x) = sign\left[\sum_{t-1}^{T}\left(\log\frac{1-\epsilon_t}{\epsilon_t}\right)\left(h_t(x)\frac{1}{2}\right)\right] \quad (8)$$

Note that for $\epsilon_t = 0.5$, the classifier t fails to participate in the combination.

It is easy to interpret the first two features for the purpose of face detection. The region of the eyes is the first feature selected by AdaBoost. This region is usually darker than the region of cheeks and nose. The second feature is the eyes which are often darker compared to the bridge of nose [8].

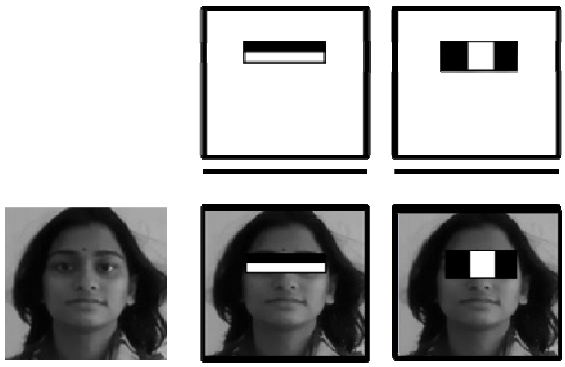


Figure 4. The first two features

3.6. Attentional cascade

The cascaded classifier is more efficient because it is smaller**.** Such a classifier rejects most of the negative subwindows and detects the majority of the positive ones. In order to discard a major portion of subwindows, simpler classifiers are used. More complex classifiers are used to gain low false positive rates.

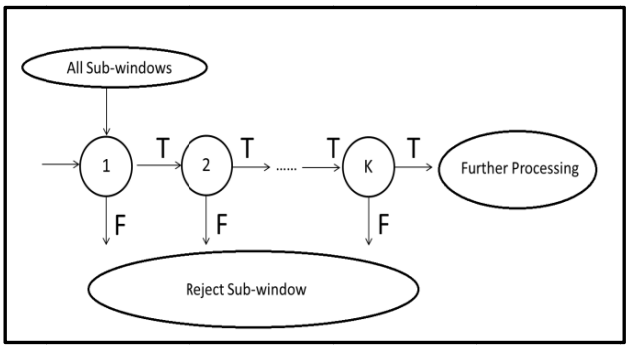


Figure 5. Schematic representation of the detection cascade (here, T: True; F: False)

For a cascade of a classifier that is trained, the false positive rate is [27]:

$$F = \prod_{i=1}^{K} f_i \quad (9)$$

where F denotes the false positive rate of the entire cas-

cade, the number of classifiers is denoted by K, and f_i is the false positive rate of the ith classifier. The rate of detection is:

$$D = \prod_{i=1}^{K} d_i \tag{10}$$

where d_i is the detection rate of the ith classifier on the examples.

The cascade training algorithm is as follows. While global false positive (FP) rate is not met, n takes the value 0. The value of n is increased by 1 so that it is $n+1$. Then the classifier is trained from n features with AdaBoost and it is evaluated on the validation set. After that, the classifier's threshold is decreased until the detection rate becomes at least d. Then again the value of n is increased until a classifier of FP rate less than f is found. Finally, the classifier is added to cascade and future classifiers are trained on false positives.

3.7. Final training

In the final training of Viola-Jones method, Adaboost training was used and 4916 faces along with 10,000 non-face sub-windows were used to train the two, five and first twenty-feature classifiers. Non-face sub-windows were gathered from the images, which did not contain faces. This was selected by sub-windows as random manner, from an image set comprising 9500 images. To train different classifiers, a number of diverse sets of non-face sub-windows were utilized to make sure that they were independent. Another reason to do so is to ensure that they did not use the same features [8].

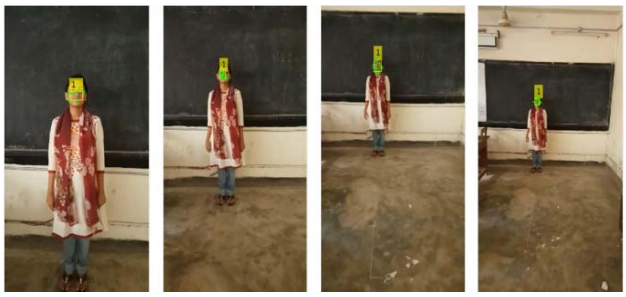
Figure 6. Frontal upright face images used for training [4]

At multiple locations and scales across the image, the final detector is scanned. Instead of scaling the image, the detector itself is scaled. Window shifting, by some number of pixels, allows the detector to be scanned. During training, in order to minimize the different effects of lighting conditions, sub-windows were variance normalized. The same measure is taken during detection. To get a single detection by combining all the overlapping detections, the detected sub-windows are post-processed [4].

4. Experimental Result and Analysis

In this paper, we have created our own data set. We have chosen 3 types of the background while making it. They are a white background, black background, noisy background. We have also varied the distance between the camera and the face. The distances between the face and the camera are as follows: 5 feet, 8 feet, 12 feet and 15 feet away from the camera. We have varied the angle of the face in the data set as well. The angels are 30-degree, 60-degree, 90-degree angle apart from the camera. We have used Xiaomi Redmi Note 4 phone which is a 13 Megapixel camera. For programming, we have used MATLAB 2015. The result of the detected face is as follows.

4.1. In black background



(a) (b) (c) (d)

Figure 7. (a) 5 feet (b) 8 feet (c) 12 feet (d) 15 feet away from the camera

4.2. In white background

(a) (b) (c) (d)

Figure 8. (a) 5 feet (b) 8 feet (c) 12 feet (d) 15 feet away from the camera

4.3. In noisy background

(a) (b) (c) (d)

Figure 9. (a) 5 feet (b) 8 feet (c) 12 feet (d) 15 feet away from the camera

It is seen that in Figure 7, the accuracy of detecting the face using Viola-Jones algorithm is 100% because there is a high contrast between the dress color of the object and the background. In Figure 8 the dress color of the object and the background is almost the same. In this case, the method shows poor accuracy when the object is 12 feet and 15 feet away from the camera. This is also to be mentioned that the camera quality is poor and the object's face is not clear in Figures 8(c) and 8(d). In Figure 9, the background is noisy and there is another object in the background. The face detection accuracy is good in Figures 9(a) and 9(b) but in the later figures, the face is not detected accurately. Another reason other than the dress color of the object is that the floor in the figures lights deep color variation.

4.4. Variation of angles

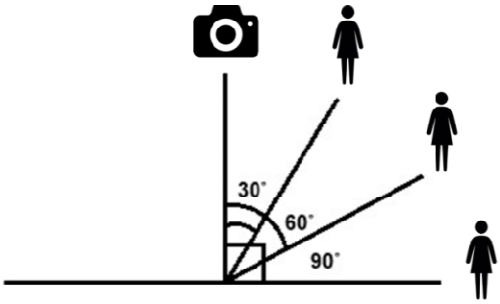
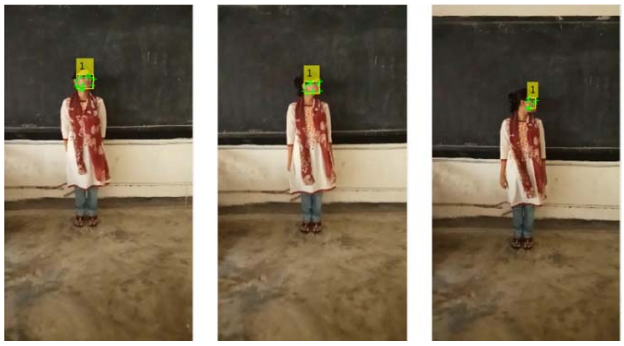


Figure 10. Variation of angles from the camera (30°, 60°, 90°)

4.5. In black background (8 feet away)



(a) (b) (c)

Figure 11. The angle between the face and the direction of the camera is (a) 30 degree (b) 60 degree (c) 90 degree.

4.6. In white background (8 feet away)

(a) (b) (c)

Figure 12. The angle between the face and the direction of the camera is (a) 30 degree (b) 60 degree (c) 90 degree.

4.7. In white background (8 feet away)

(a) (b) (c)

Figure 13. The angle between the face and the direction of the camera is (a) 30 degree (b) 60 degree (c) 90 degree.

In these figures, the object is 8 feet away but the angle between the face and the direction of the camera is 30 degree, 60 degree, and 90 degrees. In Figure 11, again the contrast with the background is high since the background color is black, that is why the detection is perfect. In the following Figure 12, the detection is also accurate in these figures because the object is 8 feet away from the camera. In the next figure, Figure 13, where the angle between the face and the direction of the camera is 90 degree, detection is not accurate.

Table 1. The experimental result of face detection at various distances and angles

Variation is distance			
	Black Background	White Background	Noisy Background
5 feet	Correct detection	Correct detection	Correct detection
8 feet	Correct detection	Correct detection	Correct detection
12 feet	Correct detection	False detection	False detection
15 feet	Correct detection	False detection	False detection
Variation in angle (8 feet away)			
	Black Background	White Background	Noisy Background
30°	Correct detection	Correct detection	Correct detection
60°	Correct detection	Correct detection	Correct detection
90°	Correct detection	Correct detection	False detection

5. Conclusion

Face detection is a difficult task under various constraints and distances. In this paper, we have explored face detection in various situations. After this research, we concluded that Viola-Jones method works well when the contrast is high between the background and the subject is high. The face detection is not accurate for noisy background. The detector can detect faces when the face of the subject creates an angle of 60 degrees or less with the camera. This performance of this method starts to decrease for the higher angle. Overall, Viola-Jones method is by far most widely used and an efficient method of detecting a face. The same method can be used to detect other objects. In that case, new training is needed.

References

[1] Hjelmas, E. and Low, B. K.: Face detection: A survey, Computer Vision and Image Understanding, Vol.83, pp.236-274, 2001.

[2] Al-Atrash, S. S. and Abuhaiba, I.: Robust Face Recognition, LAP Lambert Academic Publishing, 2012.

[3] Syeda, U. H., Zafar, Z., Islam, Z. Z., Tazwar, S. M., Rasna, M. J., Kise, K. and Ahad, M. A. R.: Visual face scanning and emotion perception analysis between autistic and typically developing children, ACM UbiComp Workshop on Mental Health and Well-Being: Sensing and Intervention, Hawaii, USA, 2017.

[4] Ahad, M. A. R.: Computer Vision and Action Recognition: A Guide for Image Processing and Computer Vision Community for Action Understanding, Springer, 2011.

[5] Ahad, M. A. R.: Motion History Images for Action Recognition and Understanding, Springer, 2012.

[6] Nishina, Y., Ahad, M. A. R., Tan, J. K., Kim, H. and Ishikawa, S.: A robust face tracking method employing color-based particle filter, Int. J. of Biomedical Soft Computing and Human Sciences, Vol.16, No.1, pp.127-134, 2010.

[7] Wang, Y. Q.: An analysis of the Viola-Jones face detection algorithm, Image Processing on Line, Vol.4, pp.128-148, 2014.

[8] Viola, P. and Jones, M.: Robust real-time face detection, International Journal of Computer Vision, Vol.57, No.2, pp.137-154, 2004.

[9] Reese, K., Zheng, Y. and Elmagharby, A.: Comparison of face detection algorithms in visible and thermal spectrums, International Conference on Advances in Computer Science and Application, 2012.

[10] Scheenstra, A., Ruifrok, A. and Veltkamp, R. C.: A survey of 3D face recognition methods, Audio- and Video-Based Biometric Person Authentication, pp.891-899, 2005.

[11] Yang, G. and Huang, T. S.: Human face detection in complex background, Pattern Recognition, 1994.

[12] Hussein, M. A., Noman, M. T. B. and Ahad, M. A. R.: A study on tiredness measurement using computer vision, Journal of the Institute of Industrial Applications Engineers, Japan, 2018.

[13] Sirohey, S. A.: Human face segmentation and identification, Technical Report CS-TR-3176, https://drum.lib.umd.edu/bitstream/handle/1903/400/CS-TR-3176.pdf;sequence=2, Univ. of Maryland, 1993.

[14] Noman, M. T. B. and Ahad, M. A. R: Mobile-based eye-blink detection performance analysis on android platform, Frontiers ICT – Human-Media Interaction, Vol.5, No.4, 2018.

[15] Irtija, N., Sami, M. and Ahad, M. A. R.: Fatigue detection using facial landmarks, The 4th Int. Symposium on Affective Science and Engineering, and the 29th Modern Artificial Intelligence and Cognitive Science Conference (ISASE-MAICS), 2018.

[16] Elahi, H. M., Islam, D., Ahmed, I., Kobashi, S. and Ahad, M. A. R.: Webcam-based accurate eye-central localization, IEEE International Conference on Robot, Vision and Signal Processing, pp.47-50, 2013.

[17] Jin, Z., Lou, Z., Yang, J. and Sun, Q.: Face detection using template matching and skin-color information, Neurocomputing, Vol.70, pp.794-800, 2007.

[18] Mutelo, R. M., Khor, L. C., Woo, W. L. and Dlay, S. S.: Two-dimensional reduction PCA: A novel approach for feature extraction, representation, and recognition, Proc. of SPIE, Vol.6060, 2006.

[19] Jee, H., Lee, K. and Pan, S.: Eye and face detection using SVM, Intelligent Sensors, Sensor Networks and Information Processing Conference, 2004.

[20] Anand, C. and Lawrance, R.: Algorithm for face recognition using HMM and SVD coefficients, Artificial Intelligent Systems and Machine Learning, Vol.5, No.3, 2013.

[21] Ahad, M. A. R.: Gait analysis: An energy image-based approach, International Journal of Intelligent Computing in Medical Sciences & Image Processing, Vol.5, No.1, pp.81-91, 2013.

[22] Hashemi, V. H. and Gharahbagh, A. A.: A novel hybrid method for face recognition based on 2d wavelet and singular value decomposition, American Journal of Networks and Communications, pp.90-94, 2015.

[23] Gao, Y. and Lee, H. J.: Viewpoint unconstrained face recognition based on affine local descriptors and probabilistic similarity, Journal of Information Processing Systems, Vol.11, pp.643-654, 2015.

[24] Sompura, M. and Gupta, V.: An efficient face recognition with ANN using hybrid feature extraction methods, International Journal of Computer Applications, Vol.117, No.17, 2015.

[25] Alshebani, Q., Premaratne, P. and Vial, P.: A hybrid feature extraction technique for face recognition,

International Proc. of Computer Science and Information Technology, pp.166-170, 2014.

[26] Wu, F.: Face recognition based on wavelet transform and regional directional weighted local binary pattern, Journal of Multimedia, Vol.9, pp.1017-1023, 2014.

[27] Viola, P. and Jones, M.: Robust real-time object detection, The 2nd International Workshop on Statistical and Computational Theories of Vision – Modeling, Learning, Computing, and Sampling, 2001.

[28] Freund, Y. and Schapire, R. E.: A decision-theoretic generalization of on-line learning and an application to boosting, Journal of Computer and System Sciences, Vol.55, No.1, pp.119-139, 1997.

[29] Mahdi, F. P., Habib, M. M., Ahad, M. A. R., Mckeever, S., Moslehuddin, A. S. M. and Vasant, P.: Face recognition-based real time system for surveillance, Intelligent Decision Technologies, Vol.11, No.1, pp.79-92, 2017.

[30] Freeman, W. T. and Adelson, E. H.: The design and use of steerable filters, IEEE Trans. Pattern Analysis and Machine Intelligence, Vol.13, No.9, pp.891-906, 1991.

[31] Kotropoulos, C. and Pitas, I.: Rule-based face detection in frontal views, Proc. of Int'l Conf. Acoustics, Speech and Signal Processing, 1997.

[32] Augusteijn, M. F. and Skujca, T. L.: Identification of human faces through texture-based feature recognition and neural network technology, Proc. of IEEE Conf. Neural Networks, pp.392-398, 1993.

[33] Rizvi, Q. M., Agarwal, B. G. and Beg, R.: A Review on Face Detection Methods, https://www.researchgate.net/publication/257338580_A_Review_on_Face_Detection_Methods, Qassim University, 2011.

[34] Ping, S. T. Y., Weng, C. H. and Lau, B.: Face detection through template matching and color segmentation, https://web.stanford.edu/class/ee368/Project_03/Project/reports/ee368group04.pdf, Stanford University.

[35] Jameel, S. K.: Face recognition system using PCA and DCT in HMM, International Journal of Advanced Research in Computer and Communication Engineering, Vol.4, No.1, 2015.

[36] Martinez, B., Valstar, M. F., Jiang, B. and Pantic, M.: Automatic analysis of facial actions: A survey, IEEE Trans. Affective Computing, 2017.

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